The basic curvature paired with a momentum section vanishes under the Poisson anchor

Hirota, Yuji, Ikeda, Noriaki

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/sigma-2025-003-comomentum-sections-and-poisson-maps-in-hami/pinned-original-source.tex (Comomentum Sections and Poisson Maps in Hamiltonian Lie Algebroids); no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: All eleven equation labels invoked inside the span (LAidentity1, Poissonidentity, lcAtosion, lcbasiccurv, Pms12, Pms13, LAIden111, identity11, P3covlocal, identity12, identity13) are defined inside the span itself; the proof is self-contained and no external numbered reference is invoked (no cite command in the span). The Hamiltonian-Lie-algebroid axioms (P1)(P2)(P3) are restated in full inside the proof, so the external definition environment (source lines 739-743, which cites [Blohmann:2018]) is standing setup only; its citation lies outside the span and is disclosed here. The proof opens with 'They are proved by using local coordinate expressions' and carries two footnotes (Einstein summation convention; the TM-connection conventions) verbatim. The substitution and differentiation steps are delivered by result with the input identities named at each step (the source's terse computational style); carried verbatim. The card face renders Lemma 3.1 by source-text counter stepping (section 3, first shared-counter environment; seed section=3, Theorem=0, equation=2).

Editorial note. The label basicms names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own Lemma environment in the article source.

Editorial note. Journal-unit assembly: card-local preamble (author theorem environments and macros verbatim), article class, no bibliography in the counted span (the span carries no \cite commands).

Editorial note. Revision r002: regenerated from the same spec after the r001 assembly crashed during packaging.

Editorial scope. the counted statement and its own proof environment only

Allowed standard prerequisites (editorial).

- A Lie algebroid (A, rho, [-,-]) over a smooth manifold M, with the local coordinate setup delivered inside the proof: coordinates x^i on M, fiber bases e_a of A and dual e^a of A*, anchor expression rho(e_a) = rho_a^i(x) partial_i, bracket [e_a, e_b] = C_{ab}^c(x) e_c, and the two Lie algebroid identities (LAidentity1) with the Cicly terms.

- A Hamiltonian Lie algebroid over a Poisson manifold: the defining axioms (P1), (P2), (P3) are restated in full inside the proof (source lines 1033-1050) as equations (Pms12), (Pms13); the definition environment itself (source lines 739-743, label HamiltonianLAS) is standing setup and cites [Blohmann:2018] externally, outside the span.

- A momentum section mu, a TM-connection nabla on A with connection 1-form omega, and the local coordinate expressions of the A-torsion, the curvature and the basic curvature (lcAtosion, lcbasiccurv), all delivered inside the proof; the Poisson condition [pi, pi]_S = 0 (Poissonidentity) likewise in-span.

Lemma 3.1. *In a Hamiltonian Lie algebroid over a Poisson manifold,* $\pi^\sharp\langle {}^A S, \mu\rangle = 0$*, where* $\langle -, -\rangle = 0$ *is the pairing of* A *and* A^**.*

Proof. They are proved by using local coordinate expressions.

For a Lie algebroid $(A, \rho, [-,-])$ over a smooth manifold M, let x^i be a local coordinate on M, $e_a \in \Gamma(A)$ be a basis of the fiber of A and $e^a \in \Gamma(A^*)$ be the dual basis on A^*. The basis of TM is denoted by $\partial_i = \frac{\partial}{\partial x^i}$. i, j, etc. are indices on M and a, b, etc. are indices on the fiber of A and A^*. Upper indices are ones for TM and A, and lower indices are ones for T^*M and A^*.[1]

Local coordinate expressions of the anchor map and the Lie bracket are $\rho(e_a) = \rho_a^i(x)\partial_i$ and $[e_a, e_b] = C_{ab}^c(x)e_c$, where $\rho_a^i(x)$ and $C_{ab}^c(x)$ are local functions. Identities of the Lie algebroid are

$$\rho_a^j \partial_j \rho_b^i - \rho_b^j \partial_j \rho_a^i = C_{ab}^c \rho_c^i, \qquad C_{ad}^e C_{bc}^d + \rho_a^i \partial_i C_{bc}^e + \mathrm{Cycl}(abc) = 0. \tag{3.3}$$

The condition of Poisson bivector field π, $[\pi, \pi]_S = 0$, is

$$\pi^{il}\partial_l \pi^{jk} + \mathrm{Cycl}(ijk) = 0. \tag{3.4}$$

A connection 1-form $\omega_a^b = \omega_{ai}^b \mathrm{d}x^i$ for the connection ∇ is introduced as $\nabla_i e_a := \omega_{ai}^b e_b$ for the basis, e_a. Local coordinate expressions of an A-torsion, a curvature and a basic curvature are given by

$$T_{ab}^c = -C_{ab}^c + \rho_a^i \omega_{bi}^c - \rho_b^i \omega_{ai}^c, \tag{3.5}$$

$$\begin{aligned} R_{ijb}^a &= \partial_i \omega_{aj}^b - \partial_j \omega_{ai}^b + \omega_{ai}^c \omega_{cj}^b - \omega_{aj}^c \omega_{ci}^b, \\ S_{iab}^c &= \nabla_i T_{ab}^c + \rho_b^j R_{ija}^c - \rho_a^j R_{ijb}^c, \\ &= -\partial_i C_{ab}^c - \omega_{di}^c C_{ab}^d + \omega_{ai}^d C_{db}^c + \omega_{bi}^d C_{ad}^c + \rho_a^j \partial_j \omega_{bi}^c - \rho_b^j \partial_j \omega_{ai}^c \\ &\quad + \partial_i \rho_a^j \omega_{bj}^c - \partial_i \rho_b^j \omega_{aj}^c + \omega_{ai}^d \rho_d^j \omega_{bj}^c - \omega_{bi}^d \rho_d^j \omega_{aj}^c, \end{aligned} \tag{3.6}$$

where the covariant derivative $\nabla_i T_{ab}^c$ is given by

$$\nabla_i T_{ab}^c = \partial_i T_{ab}^c + \omega_{di}^c T_{ab}^d - \omega_{ai}^d T_{db}^c - \omega_{bi}^d T_{ad}^c.$$

[1] Einstein summation convention is used for sums in local coordinate expressions.

Equations (P1), (P2) and (P3) in the definition of a Hamiltonian Lie algebroid are

$$\text{(P1)}\quad {}^A\nabla_a\pi^{ij} = \rho_a^k\partial_k\pi^{ij} - \partial_k\rho_a^i\pi^{kj} + \rho_b^i\omega_{ak}^b\pi^{kj} - \partial_k\rho_a^j\pi^{ik} + \rho_b^j\omega_{ak}^b\pi^{ik} = 0,$$

$$\text{(P2)}\quad \rho_a^i = \pi^{ij}\nabla_j\mu_a\big(= \pi^{ij}\big(\partial_j\mu_a - \omega_{aj}^b\mu_b\big)\big), \tag{3.7}$$

$$\text{(P3)}\quad \rho_{[a}^i\partial_i\mu_{b]} - C_{ab}^c\mu_c = -\pi^{ij}\nabla_i\mu_a\nabla_j\mu_b\big(= \pi^{ij}\big(\partial_i\mu_a - \omega_{ai}^c\mu_c\big)\big(\partial_j\mu_b - \omega_{bj}^d\mu_d\big)\big). \tag{3.8}$$

Substituting (3.7) into the identity (3.3), $\rho_a^j\partial_j\rho_b^i - \rho_b^j\partial_j\rho_a^i = C_{ab}^c\rho_c^i$, we obtain the following equation:

$$\pi^{ij}\big(-\pi^{kl}\partial_k\nabla_j\mu_a\nabla_l\mu_b - \pi^{kl}\nabla_k\mu_a\partial_l\nabla_j\mu_b - \partial_j\pi^{kl}\nabla_k\mu_a\nabla_l\mu_b - C_{ab}^c\nabla_j\mu_c\big) = 0. \tag{3.9}$$

Here, we use π is the Poisson bivector field, i.e., equation (3.4). Moreover, equation (3.9) is equivalent to

$$\pi^{ij}\big(\pi^{kl}\nabla_k\nabla_j\mu_a\nabla_l\mu_b + \pi^{kl}\nabla_k\mu_a\nabla_l\nabla_j\mu_b + \partial_j\pi^{kl}\nabla_k\mu_a\nabla_l\mu_b - T_{ab}^c\nabla_j\mu_c\big) = 0, \tag{3.10}$$

which is the covariant expression of equation (3.9).

We consider another identity as follows. Substituting (3.7) to (3.8) and using equation (3.4), we obtain

$$\pi^{ij}\nabla_j\mu_a\partial_i\mu_b - \pi^{ij}\nabla_j\mu_b\partial_i\mu_a + \pi^{ij}\nabla_i\mu_a\nabla_j\mu_b - C_{ab}^c\mu_c = 0.$$

The covariant expression of this identity is

$$\pi^{kl}\nabla_k\mu_a\nabla_l\mu_b - T_{ab}^c\mu_c = 0, \tag{3.11}$$

where equation (3.5) is used. The global form of equation (3.11) is

$$\pi(\nabla\mu(e_1), \nabla\mu(e_2)) - \big\langle\mu, {}^AT(e_1, e_2)\big\rangle = 0$$

for $e_1, e_2 \in \Gamma(A)$.

Acting the covariant derivative by ∇_j to equation (3.11), and using the identity (3.6), $S_{iab}^c = \nabla_i T_{ab}^c + \rho_b^j R_{ija}^c - \rho_a^j R_{ijb}^c$, we obtain the equation

$$\pi^{kl}\nabla_k\nabla_j\mu_a\nabla_l\mu_b + \pi^{kl}\nabla_k\mu_a\nabla_l\nabla_j\mu_b + \nabla_j\pi^{kl}\nabla_k\mu_a\nabla_l\mu_b - T_{ab}^c\nabla_j\mu_c - S_{jab}^c\mu_c = 0. \tag{3.12}$$

By taking $\pi^{ij}\times$ (3.12), we obtain the identity

$$\pi^{ij}\big(\pi^{kl}\nabla_k\nabla_j\mu_a\nabla_l\mu_b + \pi^{kl}\nabla_k\mu_a\nabla_l\nabla_j\mu_b + \nabla_j\pi^{kl}\nabla_k\mu_a\nabla_l\mu_b - T_{ab}^c\nabla_j\mu_c - S_{jab}^c\mu_c\big) = 0. \tag{3.13}$$

Compare equations (3.10) and (3.13), where note that $\nabla_j\pi^{kl} = \partial_j\pi^{kl}$ since we do not introduce a TM-connection on TM.[2] Consistency between equations (3.10) and (3.13) gives the following identity, $\pi^{ij}S_{jab}^c\mu_c = 0$, i.e., $\pi^\sharp\langle{}^AS, \mu\rangle = 0$. □

[2]We have introduced a TM-connection on A, ∇ and an A-connection on TM, ${}^A\nabla = {}^A\nabla^{\mathrm{bas}}$.